Contact Imitation Interface: Transferring Dexterous Skills by Showing How Contact Unfolds

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Project page: nus-lins-lab.github.io/CIMIWeb/

***Abstract*— Robot-free human demonstrations offer a promising route to scaling dexterous manipulation data. The key challenge is to preserve the demonstrator's natural dexterity while producing contact-grounded observations and actions that transfer to robot hands. We present Contact Imitation Interface (CIMI), a framework with contact-grounded demonstration and learning interfaces. The demonstration interface combines shared contact geometry and fingertip RGB views with contact-alignment-first retargeting and reprojection of wider scene views. The learning interface introduces a hierarchical policy in which a contact-surface planner predicts relative contact-surface trajectories and a controller coordinates wrist and finger commands to improve contact alignment. CIMI combines contact-grounded observations, retargeting, and action representations, improving mean final-stage success by at least 50 percentage points over baseline policies across six hand–task pairs on LEAP and Wuji. Both the hardware designs and software will be publicly available.**

I. Introduction

Low-cost teleoperation and portable hand-held grippers have enabled scalable demonstration collection for parallel-jaw manipulation [1, 2]. A human can fully control a gripper's low-dimensional action space through its pose and opening, using either the robot itself or a portable gripper to demonstrate manipulation. Extending this approach to dexterous hands is especially challenging for contact-sensitive tasks: differences in human and robot kinematics and contact-surface geometry make it difficult to reproduce the demonstrated sequence of contacts with the object. This calls for a demonstration interface that preserves natural human dexterity while capturing contact-grounded observations and motion that transfer to robot hands.

Existing approaches anchor demonstrations in either the human or robot embodiment. Human-centric interfaces prioritize natural manipulation with minimal constraint. Video and motion capture recover human hand motion [3–6], while instrumented gloves and wearable devices can additionally capture contact geometry or tactile interaction [7–10]. The recorded hand motion and sensory observations remain tied to the human embodiment: retargeting may not preserve the demonstrated contact evolution, and recorded observations may differ from those available during robot execution. Robot-centric interfaces instead establish compatibility through the collection hardware. Teleoperation records robot-native observations and actions [1, 11, 12], while exoskeletons and linkages mechanically map human motion to robot configurations [13–15]. This compatibility often requires robot-specific hardware or operator expertise and can restrict demonstration motion. The resulting trade-off is between demonstrator freedom and robot-side fidelity (Table I).

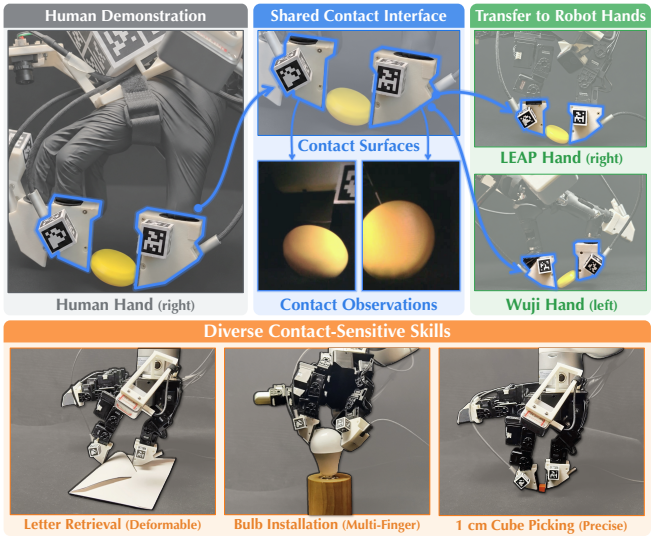


Fig. 1. **CIMI transfers contact-sensitive skills from human demonstrations to robot hands.** Shared fingertip modules provide corresponding contact surfaces and camera views for human demonstration and execution across different robot hands. Bottom: tasks involving deformable objects, multi-finger coordination, and precise contact.

To address this trade-off, we introduce the **Contact Imitation Interface (CIMI)**, which combines contact-grounded demonstration and learning interfaces:

1) We design a contact-grounded demonstration interface. Human and robot hands use fingertip modules with matching contact-surface geometry and camera placement. Each module includes an RGB camera, termed a *contact camera*, that captures local fingertip–object interactions. Cameras mounted at the wrist and thumb–index web space (*context cameras*) provide wider scene views for manipulation. Transferring demonstrations requires aligning contact motion and visual observations between human and robot hands. However, differences in hand kinematics can prevent simultaneous alignment of contact surfaces and context-camera poses. We therefore propose contact-alignment-first retargeting and context-view reprojection. By jointly optimizing wrist and finger motion, retargeting prioritizes reproducing the demonstrated contact; we retain the contact-camera

Interface	Focus	Demonstrator dexterity		Demonstration–execution interface			Cross-hand reuse
		Motion freedom	Haptic feedback	Contact geometry	Contact observations	Action mapping	
Bare-hand video [5, 6, 16–22]	Human	Unconstrained	Direct touch	Mismatched human contact surfaces	None	Hand-pose retargeting	✓
Sensing glove [7–9, 23]	Human	Near-free	Through glove	Glove contact surfaces; matching varies	Human/shared tactile (optional)	Glove-state retargeting	✓
Teleoperation [11, 12, 24–33]	Robot	Controller/ robot-limited	Remote/optional	Matched robot contact surfaces	Robot sensing (optional)	Native robot commands	✓/×
Fingertip exoskeleton [13, 15, 34]	Robot	Exoskeleton-limited	Through contact surfaces	Matched robot contact surfaces	Matched tactile (optional)	Robot-aligned device state	×
Passive linkage [14, 35]	Robot	Linkage-limited	Through linkage	Robot-aligned contact surfaces	Robot-aligned tactile (optional)	Direct joint transfer	×
CIMI (Ours)	**Contact**	**Near-free**	**Through contact surfaces**	**Matched shared contact surfaces**	**Matched contact cameras**	**Contact-surface retargeting**	✓

TABLE I: Demonstration interfaces: dexterity, contact geometry and observations relative to execution, and cross-hand reuse. Mixed entries vary across cited systems.

images and reproject context views to the robot viewpoints.

2) We develop a contact-grounded learning interface. Policies commonly predict relative wrist motion and finger-joint commands [13]. In contact-sensitive tasks with high-degree-of-freedom hands, we observe that this representation can lead to poor wrist–finger coordination and contact misalignment. We propose a hierarchical policy where a contact-surface planner predicts relative contact-surface trajectories from visual and proprioceptive inputs. A contact-surface controller converts these trajectories into coordinated wrist and finger commands.

By centering both demonstration and learning on contact surfaces, CIMI enables contact-sensitive dexterous skills to be learned from natural human demonstrations and reproduced across different robot hands. Experiments on LEAP and Wuji cover contact-sensitive tasks including grasping a 1 cm cube, wedging and lifting a flat coin, retrieving a letter, and screwing in a bulb. Across six hand–task pairs, CIMI improves mean success by at least 50 percentage points over baseline policies. Videos are available at https://nus-lins-lab.github.io/CIMIWeb/.

II. Related Work

A. Demonstration Interfaces for Dexterous Manipulation

Human-centric interfaces capture manipulation directly from the human hand. Bare-hand approaches capture or reconstruct human-hand motion—and, when available, object motion—from RGB or egocentric videos [5, 6, 16, 18–21]. Motion-capture systems provide more reliable hand and wrist trajectories [3, 4], while instrumented gloves additionally capture touch, tactile signals, or contact-surface trajectories [7–9, 23]. For example, DexCap [3] combines hand motion capture with scene point clouds and fingertip-position retargeting. CIMI provides contact-grounded observations from matched fingertip cameras. Its contact-alignment-first retargeting improves contact alignment and task success over DexCap-style retargeting (Table III).

Robot-centric interfaces instead collect demonstrations through robot-compatible hardware. Teleoperation maps human input to an actuated robot hand and directly records robot actions and observations [11, 12, 24–32, 36–38]. Fingertip exoskeletons constrain or measure human motion in a robot-oriented space [13, 34], while linkage-based exoskeletons and proxy hands mechanically align human motion with robot kinematics [14, 35]. For example, DexUMI [13] maps exoskeleton motion directly to robot joints, constraining demonstrations to the target hand's kinematics: 1–2 active DoFs per finger on Inspire and 2–3 on XHand. CIMI supports four active DoFs per finger without imposing robot-specific finger kinematics on the demonstrator. Its contact-grounded planner and controller improve success over direct wrist–joint prediction on our contact-sensitive tasks (Table II).

B. Learning Dexterous Manipulation from Human Demonstrations

We focus on learning robot policies from demonstrations captured through human hand motion and sensing, rather than transferring fully specified hand–object motion-capture trajectories [39, 40]. The key challenge is aligning human demonstrations with robot execution in both action and observation spaces. Existing methods address this challenge through explicit transformation, cross-embodiment policy learning, or both. For action alignment, human wrist, hand, or fingertip motion is retargeted into robot joint or end-effector commands [3, 5, 6, 9, 13, 41]. For observation alignment, demonstration videos can be transformed through human-hand segmentation, background inpainting, and robot-hand compositing, or matched sensing modalities can be used during both demonstration and execution [8, 9, 13]. Learning-based approaches use pretraining, co-training, or representation alignment to ground human-derived representations in robot sensing and control [4, 5, 18, 21].

CIMI uses matched fingertip contact surfaces as a shared cross-embodiment interface: tracked contact-surface trajectories supervise robot actions, while local sensing aligns policy observations, without imposing robot-specific hand kinematics on the demonstrator. A contact-surface planner and controller use this shared representation to separate contact-surface trajectory planning from wrist and finger control.

III. Contact Imitation Interface

CIMI uses contact surfaces to connect human demonstration and robot learning (Fig. 2). The demonstration

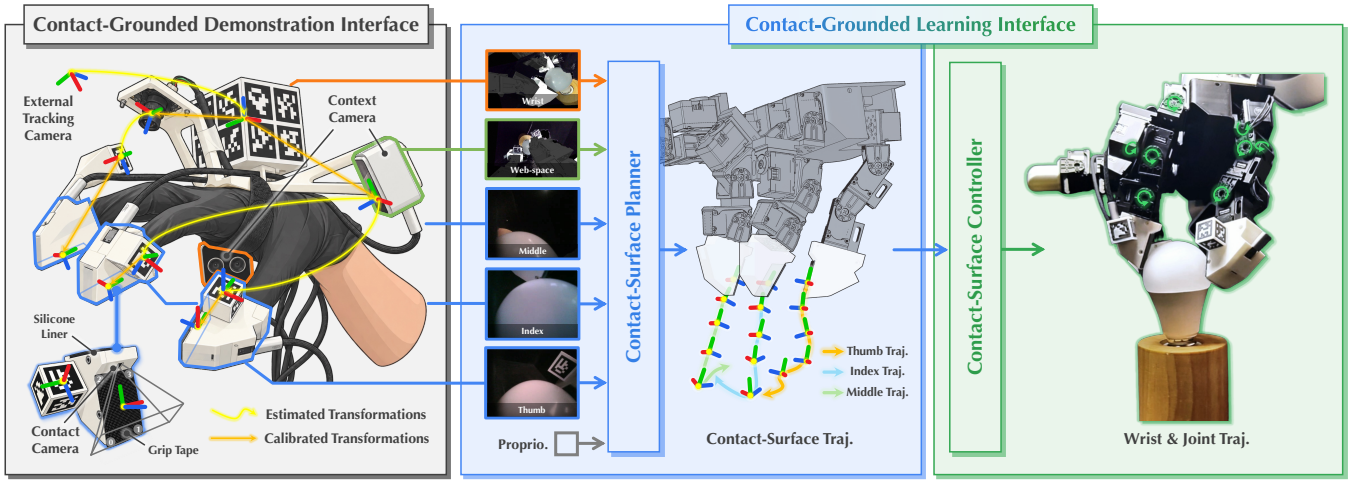


Fig. 2. **Contact connects demonstration and learning in CIMI.** The demonstration interface captures contact-surface trajectories and observations, then constructs robot demonstrations through contact-alignment-first retargeting and context-view reprojection. The learning interface predicts relative contact-surface trajectories and realizes them through coordinated wrist and finger control.

interface transfers contact motion and visual observations to robot hands (Section III-A). The learning interface predicts relative contact-surface trajectories and converts them into coordinated wrist and finger commands (Section III-B).

A. Contact-Grounded Demonstration Interface

The demonstration interface converts human demonstrations into contact-grounded robot motion targets and observations. We achieve this through four design choices. Shared fingertip hardware establishes matching contact-surface geometry and camera placement (**D1**), while visual tracking recovers demonstrated contact-surface trajectories (**D2**). Contact-alignment-first retargeting jointly optimizes wrist and finger motion to reproduce these trajectories (**D3**). We retain contact-camera views and reproject context-camera views to match the retargeted motion (**D4**).

D1. Contact-Grounded Capture Hardware. CIMI captures contact-surface trajectories and observations using cameras and printed fiducials. Human and robot fingertip modules share contact-surface geometry and camera placement. Each RGB camera (*contact camera*) views the local contact region from the fingertip root, tangentially to the surface. Two Gemini 305 stereo cameras view the wider scene (*context cameras*) from the wrist and thumb–index web space. We record nine synchronized streams at 10 Hz: three contact-camera views (thumb, index, and middle fingers), four context views, and middle-finger and external tracking views. Hardware specifications, fabrication details, and a bill of materials are provided in Supp. Sec. I-A.

D2. Visual Contact-Surface Tracking. We recover contact-surface trajectories by composing estimated poses and calibrated transforms (Fig. 2). We use AprilTag markers [42] on AprilCube targets [43], with DeepTag [44] providing dense grid keypoints. We jointly fit a target pose to keypoints across visible tags in one multi-tag Perspective-n-Point (PnP) problem [45]. Visible tags provide redundant constraints when other tags are occluded. Frame definitions and estimator details are provided in Supp. Sec. I-B.

D3. Contact-Alignment-First Retargeting. Our retargeting prioritizes contact-surface alignment over context-camera poses. Fixing the wrist trajectory to align human and robot context-camera poses (*context-alignment-first*) leaves only finger joints to reduce contact error (Fig. 3a). Following HoMMI's relaxation of camera-pose constraints [46], we free the wrist to improve contact alignment and correct the resulting viewpoint mismatch through reprojection (Fig. 3b; **D4**). Table III evaluates this choice. Human contact-surface poses and retargeted robot trajectories share the external tracking-camera frame. We optimize wrist poses $\mathbf{X}_t$ and finger joints $\mathbf{q}_t$ to match robot corners $\mathrm{FK}_{i,\ell}(\mathbf{X}_t, \mathbf{q}_t)$ to tracked human corners $\mathbf{p}^H_{i,\ell,t}$, where FK denotes forward kinematics. Matching four corners ℓ per finger i (Fig. 2, left) aligns contact-surface position and orientation:

$$\{\mathbf{X}^*_t, \mathbf{q}^*_t\}_{t=1}^T = \underset{\{\mathbf{X}_t, \mathbf{q}_t\}_{t=1}^T}{\operatorname{argmin}} \Big[\sum_{t,i,\ell} \|\mathrm{FK}_{i,\ell}(\mathbf{X}_t, \mathbf{q}_t) - \mathbf{p}^H_{i,\ell,t}\|_2^2 + \mathcal{R} \Big].$$

The smoothness regularizer $\mathcal{R}$ penalizes changes in wrist translation, rotation, and finger joints between valid neighboring frames. We initialize the optimization by aligning human and robot context-camera poses. We solve the retargeting optimization using PyRoki [47], subject to the robot's finger-joint limits. From the retargeted trajectory, we derive relative contact-surface trajectories to supervise the planner (Section III-B). The wrist and joint trajectories also provide the basis for motion augmentation and controller training. The retargeting objective and optimization settings are detailed in Supp. Sec. I-C.

D4. Contact Observations and Context-View Reprojection. We construct demonstration observations to align with the contact and context views available during robot execution. Shared contact-surface geometry and camera placement give corresponding human and robot cameras the same nominal viewpoint relative to the local contact surface, so we retain the recorded contact-camera images. Context-camera viewpoints change with the retargeted wrist motion (**D3**), so we convert their recorded views through reprojection

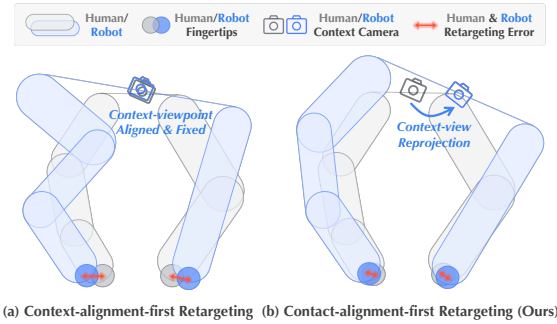


Fig. 3. **Retargeting priorities.** (a) Context-alignment-first retargeting matches context-camera poses but can leave contact surfaces misaligned. (b) Contact-alignment-first retargeting adjusts the wrist to improve contact alignment; reprojection corrects the viewpoint mismatch. Gray: human; blue: robot.

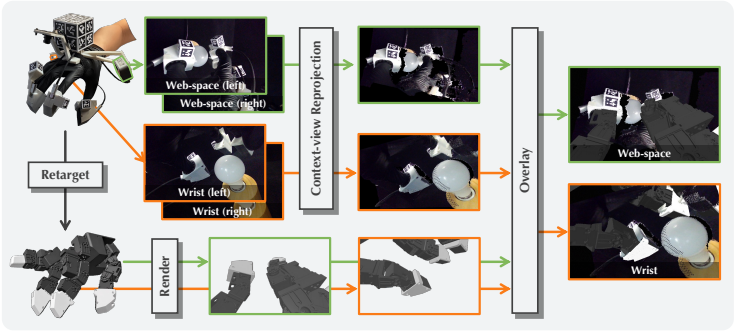


Fig. 4. **Context-view reprojection and robot-geometry overlay.** Recorded web-space (green) and wrist (orange) stereo images are reprojected to camera poses determined by retargeting. Robot geometry is rendered at the retargeted configuration and overlaid on these views to produce RGB training observations from the robot perspective. This conversion is performed offline.

and robot-geometry overlay to match the robot perspective (Fig. 4). We optimize robot context-camera mounts using collected demonstrations (details in Supp. Sec. I-D). For each stereo pair, we estimate depth with Fast-FoundationStereo [48] and back-project the RGB images and estimated depth into a colored point cloud. We transform this cloud into the robot camera frame determined by these mounts and the retargeted wrist and hand configuration, reproject it onto the image plane, and composite it with the robot geometry using nearest-depth visibility. The resulting RGB context views, recorded contact-camera images, and retargeted wrist–hand states form synchronized demonstration samples for policy training. Demonstration conversion is performed offline. At execution, the policy receives contact RGB images and preprocessed context RGB views; depth is used only for context-image preprocessing.

B. Contact-Grounded Learning Interface

The learning interface separates contact-surface trajectory planning from wrist–finger control to improve coordination (**O3**). Direct command prediction [13] requires a visual policy to learn both contact motion and its kinematic realization. We instead represent actions as relative contact-surface trajectories (**L1**), train a visual planner to predict them from demonstrations (**L2**), and train a separate controller to convert them into wrist and finger commands using augmented synthetic motion data (**L3**).

L1. Relative Contact-Surface Trajectories as Actions. The planner predicts how each contact surface should move to establish and maintain contact with the object. We express this motion relative to the contact surface's current pose, aligning actions with observations from the contact camera on the same fingertip module.

Let F be the number of controlled fingers, and let $\mathrm{FK}_i(\mathbf{q}_t)$ denote the pose of contact surface i in the wrist frame. Its pose in a retargeted demonstration is $\mathbf{Y}_{i,t} = \mathbf{X}_t^* \mathrm{FK}_i(\mathbf{q}_t^*)$. For each contact surface, all future poses in a chunk use its frame at prediction time t as the reference, following the relative-trajectory convention of [2]:

$$\Delta\mathbf{T}_{i,t,h} = \mathbf{Y}_{i,t}^{-1}\mathbf{Y}_{i,t+h}.$$

Relative actions are invariant to fixed changes of the global frame's origin and orientation. Each pose uses three translation coordinates and six rotation coordinates (the rotation matrix's first two rows). Concatenating them across F fingers gives $\mathbf{u}_{t,h} \in \mathbb{R}^{9F}$.

L2. Contact-Surface Planner. A frozen C-RADIOv3-B backbone [49] extracts a summary token per contact or context view. View-specific projections and an MLP fuse the tokens into $\mathbf{z}_t^{\mathrm{vis}}$. The proprioceptive input $\mathbf{s}_t^{\mathrm{prop}}$ encodes thumb-relative contact-surface poses, exposing inter-finger geometry for contact coordination. This input outperforms the tested alternatives in Table IV. Given $\mathbf{o}_t = (\mathbf{z}_t^{\mathrm{vis}}, \mathbf{s}_t^{\mathrm{prop}})$, we train a flow-matching planner $\pi_\theta(\mathbf{U}_t \mid \mathbf{o}_t)$ [50] on demonstrated contact-surface trajectories. It outputs an H-step relative contact-surface trajectory $\widehat{\mathbf{U}}_t = [\widehat{\mathbf{u}}_{t,1}, \ldots, \widehat{\mathbf{u}}_{t,H}]$. We use visual augmentation during planner training to reduce sensitivity to reprojection artifacts, the appearance gap between rendered and real robot images, and photometric differences between demonstration and robot cameras. Training details are in Supp. Sec. II-A.

L3. Contact-Surface Controller. The MLP controller C_ϕ maps current joint values $\mathbf{q}_t$ and the predicted contact-surface trajectory $\widehat{\mathbf{U}}_t$ to future wrist poses $\widehat{\Delta\mathbf{X}}_{t,h}$ relative to the current wrist pose and absolute finger-joint targets $\widehat{\mathbf{q}}_{t,h}$:

$$\left[(\widehat{\Delta\mathbf{X}}_{t,h}, \widehat{\mathbf{q}}_{t,h})\right]_{h=1}^{H} = C_\phi(\mathbf{q}_t, \widehat{\mathbf{U}}_t).$$

To cover predicted contact-surface trajectories beyond recorded demonstrations, we augment retargeted trajectories from **D3** with smooth wrist and joint perturbations. Forward kinematics provides paired contact-surface trajectories and wrist–joint commands. We supervise the controller with wrist–joint command regression and contact-surface alignment measured through forward kinematics. Input construction, augmentation, and loss details are provided in Supp. Sec. II-B.

At execution, the controller converts the trajectory into commands in one forward pass. We use $H = 16$ and execute an 8-step prefix, then replan from the latest observation. We reuse the controller across tasks with the same robot hand and controlled fingers.

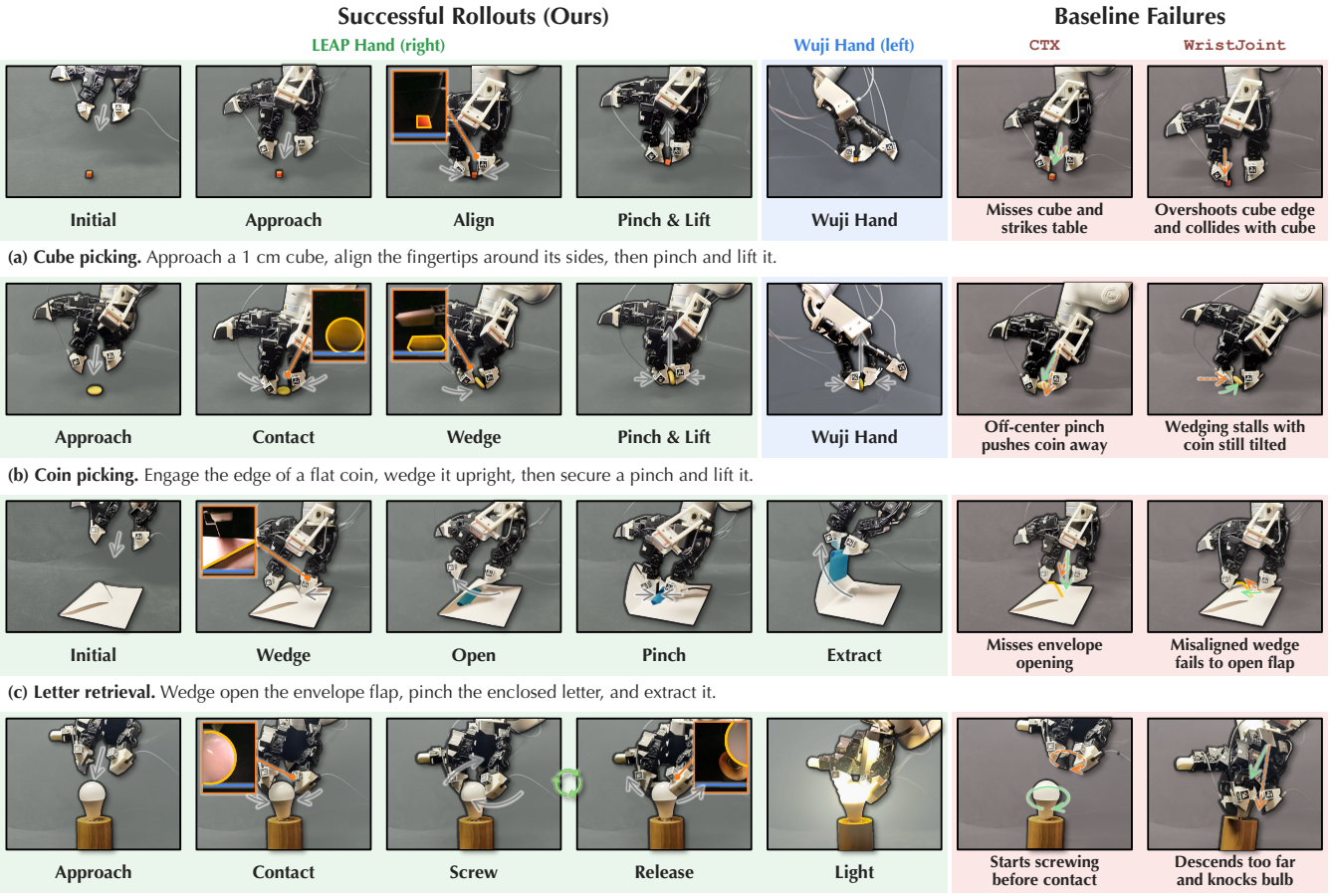


Fig. 5. **Real-world rollouts and failure cases.** Orange-bordered insets (▭) show contact-camera views; within each inset, blue lines (—) mark the contact surface and yellow outlines (—) mark object boundaries. Green (▭) and blue (▭) backgrounds show successful LEAP and Wuji rollouts, respectively; red (▭) marks **`CTX`** and **`WristJoint`** failures.

Policy	Visual	LEAP Hand								Wuji Hand				Mean
		Cube		Coin		Letter		Bulb		Cube		Coin		
		Align	Lift	Wedge	Lift	Open	Lift	Screw	Light	Align	Lift	Wedge	Lift	
`WristJoint`	`CTX`	0	0	0	0	0	0	35	5	0	0	0	0	0.8
`ContactSurface`	`CTX`	0	0	0	0	0	0	50	10	0	0	0	0	1.7
`WristJoint`	`CTX+CC`	35	20	50	35	25	15	85	70	30	15	40	30	30.8
`ContactSurface`	`CTX+CC`	**85**	**75**	**95**	**85**	**90**	**80**	**95**	**90**	**75**	**70**	**90**	**85**	**80.8**

TABLE II: Policy and observation comparison on LEAP and Wuji hands. SR in %; 20 rollouts per method, hand, and task. Mean SR averages final-stage SR equally over all six hand–task pairs.

IV. Experiments

We evaluate whether CIMI's contact-grounded demonstration and learning interfaces enable robot hands to learn contact-sensitive dexterous manipulation from human demonstrations. The experiments answer six questions: **Q1: Learning across robot hands.** Can CIMI enable different robot hands to learn contact-sensitive skills from the same human demonstrations? **Q2: Contact-grounded observations.** Do contact-grounded observations improve manipulation performance over context vision alone? **Q3: Contact-grounded policy.** Do hierarchical contact-surface planning and control improve manipulation over direct wrist and joint prediction? **Q4: Contact-surface alignment.** Is accurate contact-surface alignment important for contact-sensitive manipulation? **Q5: Collection efficiency.** How fast can CIMI collect demonstrations compared with bare-hand demonstration and teleoperation? **Q6: Design ablations.** Which design choices contribute most to reliable manipulation with CIMI?

A. Q1–Q3: Contact-Grounded Policy Evaluation

Tasks and Evaluation Protocol. Fig. 5 shows LEAP and Wuji rollouts. LEAP tasks comprise cube picking (146 training demos), coin picking (187 demos), letter retrieval (183 demos), and bulb installation (103 demos); Wuji tasks comprise cube picking (same 146 demos) and coin picking

(same 187 demos). These tasks test fingertip alignment, wedging, envelope opening, and repeated bulb regrasping. We report stage-wise success rates (SR) over 20 rollouts per method, hand, and task; mean SR averages final-stage success across the six hand–task pairs (Table II).

Compared Policies. Table II crosses two visual inputs—**`CTX`** (task-specific context-camera views) and **`CTX+CC`** (the same context views plus contact-camera images)—with two policies: **`WristJoint`** (direct prediction of relative wrist motion $(\mathbf{X}_t^*)^{-1}\mathbf{X}_{t+h}^*$ and joint changes $\mathbf{q}_{t+h}^* - \mathbf{q}_t^*$, following DexUMI [13]) and **`ContactSurface`** (our contact-grounded policy). Context views include wrist RGB and, depending on the task, web-space RGB; they are matched between **`CTX`** and **`CTX+CC`**. Supp. Sec. II-E specifies the camera views for each task. All variants for a given hand use the same demonstrations and retargeted trajectories. Within each visual setting, policy comparisons match visual and proprioceptive inputs, flow-model width, and planner training budget. For cube and coin picking, we retarget the same human recordings to each hand and train separate policies.

O1: Human demonstrations support learning on both hands. CIMI achieves 80.8% mean final-stage SR across all six hand–task pairs with **`ContactSurface`** and **`CTX+CC`** (Table II). For the two tasks shared across hands, cube lifting succeeds in 75% of LEAP rollouts and 70% of Wuji rollouts, while coin lifting reaches 85% on both. LEAP also achieves 80% on letter retrieval and 90% on bulb installation. The successful rollouts in Fig. 5 show policies learned through CIMI reproducing demonstrated contact sequences on different robot hands.

O2: Contact-camera feedback improves contact establishment. Adding contact-camera views raises mean final-stage SR from 0.8% to 30.8% for **`WristJoint`** and from 1.7% to 80.8% for **`ContactSurface`** (Table II). With context vision alone, both policies score 0% even at the first stage of cube, coin, and letter tasks. The context-only examples in Fig. 5 show errors at contact establishment: missing the cube and striking the table, pushing the coin away with an off-center pinch, missing the envelope opening, and starting to screw before contacting the bulb. These examples illustrate the local placement and engagement cues that contact-camera views provide beyond context vision.

O3: Contact-surface planning better uses contact feedback. With **`CTX+CC`**, **`ContactSurface`** exceeds **`WristJoint`** by 50.0 percentage points in mean final-stage SR (80.8% versus 30.8%). Gains span all six hand–task pairs, ranging from 20 points on LEAP bulb to 65 on LEAP letter, with 55-point gains on both Wuji tasks. The **`WristJoint`** examples in Fig. 5 show residual motion errors: overshooting the cube edge, stalling while wedging the coin, misaligning with the envelope flap, and descending too far onto the bulb. This pattern supports the design in Section III-B: plan motion in the contact-surface frame and use a separate controller to coordinate wrist and finger commands. The comparison evaluates this combined design; Q6 examines individual input and augmentation choices.

Retargeting		Coin			Bulb		
Target	Priority	Error (mm)↓	SR (%)↑		Error (mm)↓	SR (%)↑	
			Wedge	Lift		Screw	Light
Centers	Context	15.37	0	0	7.85	10	0
Corners	Context	14.67	0	0	5.72	25	0
Centers	Contact	9.31	0	0	4.36	50	20
Corners	**Contact**	**0.51**	**95**	**85**	**1.09**	**95**	**90**

TABLE III: Retargeting ablation on LEAP. Priority denotes context- or contact-alignment-first. Gray: DexCap-style [3]; blue: ours. Error is the mean contact-corner distance over valid frames. SR uses 20 rollouts per variant and task; our SRs are from Table II.

B. Q4: Contact-Surface Alignment

Tasks and Evaluation Protocol. We compare center and corner matching under context- and contact-alignment-first retargeting on LEAP coin picking and bulb installation (Table III). Center matching aligns the centroid of each surface's four corners without constraining orientation. *Context-alignment-first* fixes the wrist to match context-camera poses; *contact-alignment-first* jointly optimizes the wrist and finger joints to align contact surfaces. All variants use the same human recordings. We measure mean Euclidean error (mm) between retargeted and demonstrated surface corners (four per finger) (**D3**). For each task, we retrain planners for all variants with identical data splits, architectures, training budgets, and controllers, then evaluate each in 20 rollouts.

O4: Contact priority and corner matching jointly improve contact transfer. All three alternative retargeting variants yield 0% Coin Wedge SR but 10–50% Bulb Screw SR (Table III). This contrast may reflect greater tolerance to contact error during bulb engagement than during coin-edge wedging. Completing bulb installation remains sensitive to alignment: both context-alignment-first variants have 0% Light SR. With contact priority, replacing center matching with corner matching reduces bulb contact-corner error from 4.36 to 1.09 mm and raises Light SR from 20% to 90%.

C. Q5: Collection Efficiency

Tasks and Evaluation Protocol. We compare CIMI, bare hands, and teleoperation by counting successful demonstrations collected by the same operator in 15 minutes (Fig. 6). We evaluate this collection throughput (CT) on coin picking and bulb installation, using the same task success criteria across interfaces. Teleoperation combines Manus hand tracking, Vive wrist tracking, and GeoRT [36] hand retargeting. The timed protocol includes failed attempts, resets, and object randomization.

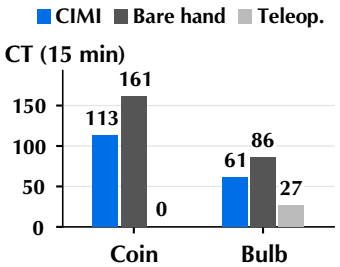


Fig. 6. **15-minute collection throughput on coin picking and bulb installation.**

Variant	Coin		Bulb		Mean SR (%)↑
	Wedge	Lift	Screw	Light	
Context-view conversion					
No overlay	90	80	75	45	62.5
No reprojection, no overlay	75	50	55	10	30.0
Planner proprioceptive input					
No planner proprioception [13]	90	85	10	0	42.5
Joint values [1, 3]	70	55	80	50	52.5
Relative wrist + joints [2, 46]	60	40	30	5	22.5
Augmentation					
No planner visual augmentation	0	0	15	0	0.0
Task-only controller, no motion aug.	90	70	80	35	52.5
Ours	**95**	**85**	**95**	**90**	**87.5**

TABLE IV: Design ablations on LEAP. Stage-wise SR in %; 20 rollouts per variant and task. All variants use our `ContactSurface` + `CTX+CC` framework with the listed changes. Citations denote related input styles adapted to CIMI. Our SRs are from Table II. Mean SR averages Coin Lift and Bulb Light.

O5: CIMI supports efficient collection of transferable demonstrations. CIMI retains 70% of bare-hand throughput on coin picking and 71% on bulb installation (Fig. 6). It achieves 2.3× teleoperation throughput on bulb, while teleoperation yields no successful coin demonstrations. Together with Table II, these results support efficient demonstration collection for contact-sensitive robot learning.

D. Q6: Design Ablations

Tasks and Evaluation Protocol. We ablate context-view conversion, planner proprioception, and visual and motion augmentation on LEAP coin picking and bulb installation (Table IV). Each variant changes the listed components of `ContactSurface` with `CTX+CC`, using the same stage criteria and 20-rollout evaluation protocol.

O6: Context-view conversion improves contact control. Removing the robot-geometry overlay while retaining reprojection lowers mean SR from 87.5% to 62.5%; using the original human context views without either operation lowers it further to 30.0% (Table IV). In these rollouts, overall task motion remains reasonable, but contact control becomes less precise, despite unchanged contact-camera inputs. The observed errors suggest that reducing differences in context-camera viewpoints and robot appearance helps refine contact control.

O7: Thumb-relative proprioception improves planner performance. Thumb-relative contact-surface poses yield 87.5% mean SR, compared with 52.5% for joint-value inputs (style of [1, 3]). With joint values, coin rollouts show slightly looser fingertip contact, and Coin Lift SR falls from 85% to 55%. Removing planner proprioception (style of [13]) preserves 85% Coin Lift SR but reduces Bulb Light SR from 90% to 0%. In bulb rollouts, relative-pose errors accumulate over successive motions, and the middle finger drifts away from the bulb. The relative proprioception variant (style of [2, 46]) uses the wrist pose relative to the previous frame and inter-frame joint changes. This variant yields 22.5% mean SR and can stall during bulb screwing, predicting little or no motion.

O8: Visual augmentation and broader controller training improve success. Without planner visual augmentation, final-stage SR is 0% on both tasks, although Bulb Screw reaches 15%. Training the controller only on the target task's retargeted trajectories, without synthetic motion augmentation, reduces mean SR from 87.5% to 52.5%: Coin Lift falls from 85% to 70%, and Bulb Light from 90% to 35%. The larger drop on bulb installation suggests that this task benefits especially from broader controller training. This comparison changes both motion augmentation and training-data coverage.

V. Conclusion

CIMI connects natural human demonstrations to robot hands through contact-grounded observations, retargeting, and action representations. Across six hand–task pairs on LEAP and Wuji, it improves mean success by at least 50 percentage points over baseline policies. These results support shared contact surfaces as a practical interface for transferring contact-sensitive skills across different hand kinematics. We will release the hardware designs and software to support broader robot-free demonstration collection. Extending CIMI to bimanual and mobile manipulation offers a path toward transferring a wider range of everyday skills.

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Contact Imitation Interface: Transferring Dexterous Skills by Showing How Contact Unfolds

Supplementary Material

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SUPPLEMENTARY CONTENTS

I Demonstration Interface 9
I-A Contact-Grounded Capture Hardware . . . 9
I-B Visual Contact-Surface Tracking 10
I-C Contact-Surface Retargeting 11
I-D Context-Camera Mount Optimization . . . 11
I-E Processing Efficiency and Data Quality . . 12
II Learning Interface 12
II-A Contact-Surface Planner 12
II-B Contact-Surface Controller 13
II-C Execution and Reuse 14
II-D Discussion on Next-State Action Targets . 14
II-E Policy Camera Inputs and Preprocessing . 15
II-F Training and Inference Efficiency 15
References 15

I. DEMONSTRATION INTERFACE

This section expands the capture hardware (D1), visual tracking (D2), contact-surface retargeting (D3), and context-camera mounting used for observation conversion (D4) in the main paper.

A. Contact-Grounded Capture Hardware

This section describes the hardware design and fabrication of the demonstration interface in D1 of the main paper. The design combines matched fingertip modules, wider-view stereo cameras, and dedicated tracking views while keeping the custom mechanical components straightforward to reproduce.

1) Why Visual Tracking?: **Direct contact-surface calibration.** Visual tracking directly measures our rigid contact modules using known target geometry and calibrated target-to-surface transforms. A glove-plus-tracker system instead requires additional registration from its calibrated hand model to these contact surfaces [1, 2].

Contact-camera compatibility. In our tests with MANUS Metagloves Pro, MANUS Core 3, and SDK access, simultaneous contact-camera operation could degrade image quality or shift the estimated hand pose, consistent with electromagnetic interference; MANUS also warns of this sensitivity [2].

Lower hardware cost. The vendor lists Metagloves Pro at EUR 4,500 [3], while our capture-interface hardware totals $778.95 (Table S1). The listed glove price does not establish the cost of our tested software/SDK configuration.

2) Hardware Design Details: **Matched fingertip modules.** Each human or robot fingertip module uses one HBVCAM-2307-FPC82 V11 USB camera. The camera measures approximately $8.6 \times 8.6 \times 5.4$ mm and has a 120° field of view. It is mounted at the fingertip root and oriented tangentially toward the local contact region. The human and robot variants preserve the same nominal contact-surface geometry and camera placement, with hand-specific 3D-printed mounting structures. A silicone sleeve provides compliant contact without obstructing the camera view. We apply one layer of friction tape on the human contact surface and two on the robot surface; the additional robot-side thickness supports contact compression, as discussed in Sec. II-D.

Context and tracking cameras. Two Orbbec Gemini 305 stereo cameras are mounted at the wrist and near the thumb–index web space. Together they provide the four context-image streams used for reconstruction and viewpoint conversion. One DCXG500 RGB camera tracks the middle-finger fiducial, and an external DCXG500 camera tracks the dorsal fiducial on the back of the hand. The fingertip targets, dorsal target, and tracking-camera mounts are rigidly attached so that their fixed transformations can be calibrated once and reused across recordings.

Fabrication and assembly. We first 3D print the fingertip bodies, camera holders, hand-specific adapters, tracking-camera mounts, and fiducial supports. We then install one contact camera and one silicone sleeve on each fingertip module, attach the AprilCube targets, and mount the modules to the human interface or robot hand. Finally, we install the two Gemini 305 cameras and the auxiliary tracking cameras, calibrate camera intrinsics and the fixed transformations in Fig. S1, and verify all nine streams at the 10 Hz recording rate. The printed geometry fixes the nominal camera–contact-surface placement, which is shared between human demonstration and robot execution.

3) Bill of Materials: Table S1 lists the principal components for one three-finger demonstration interface in

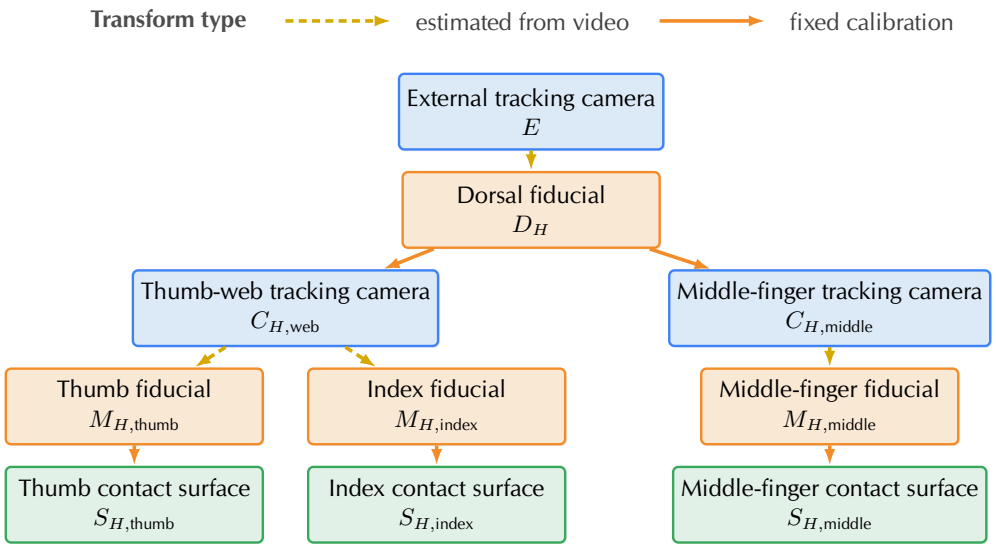


Fig. S1. **Transformation tree for visual contact-surface tracking.** Each edge specifies the child frame's pose in its parent frame. Dashed yellow edges are estimated from videos; solid orange edges are fixed mounting calibrations.

USD; the contact-camera price follows the FingerEye BOM. Printed parts, silicone sleeves, friction tape, and printed fiducials have a combined estimated material cost of $7.00. Costs exclude labor, printer depreciation, cables, and the host computer.

Component	Qty.	Cost (USD)
HBVCAM-2307-FPC82 V11 contact camera	3	$28.25 each
Orbbec Gemini 305 stereo camera	2	$259.18 each
DCXG500 RGB tracking camera (middle-finger and external views)	2	$84.42 each
Fabrication materials (printed parts, silicone sleeves, friction tape, and fiducials)	1 set	$7.00 (est.)
Estimated hardware total		$778.95

TABLE S1: Bill of materials for one three-finger demonstration interface. The total covers three contact cameras, two Gemini 305 cameras, two auxiliary tracking cameras, and the combined fabrication-material estimate.

B. Visual Contact-Surface Tracking

This section details the visual estimator and calibrated transformation tree used to extract contact-surface trajectories in D2 of the main paper.

1) Fiducial Targets and Tracking Frames: We use AprilTag markers [4] on AprilCube targets [5] and detect their dense grid keypoints with DeepTag [6]. Each fingertip target is rigidly attached to its contact module. A dorsal fiducial on the back of the hand provides a moving reference that connects the hand-mounted tracking cameras to the external tracking-camera frame.

Let ${}^{A}\mathbf{T}_{t}^{B} \in \mathrm{SE}(3)$ map coordinates from frame B to frame A at time t. The subscript H denotes the human demonstration interface. We use E for the fixed external tracking-camera frame, D_H for the moving dorsal fiducial frame, $M_{H,i}$ for fingertip i's target frame, and $S_{H,i}$ for its contact-surface frame. The index $j(i)$ identifies the hand-mounted tracking camera that observes fingertip i, with frame $C_{H,j(i)}$. The thumb and index targets share a thumb-web tracking view, while the middle finger uses a separate tracking view:

$$j(\text{thumb}) = j(\text{index}) = \text{web},$$
$$j(\text{middle}) = \text{middle}.$$

These tracking views estimate target poses; the contact cameras described in D1 provide the policy's contact observations.

2) Full Contact-Surface Transformation Tree: Figure S1 shows the complete tree for the three-finger capture configuration. Each contact-surface pose in the external camera frame is obtained by composing two tracked poses and two fixed mounting transforms:

$${}^{E}\mathbf{T}_{t}^{S_{H,i}} = {}^{E}\mathbf{T}_{t}^{D_H}\, {}^{D_H}\mathbf{T}^{C_{H,j(i)}}\, {}^{C_{H,j(i)}}\mathbf{T}_{t}^{M_{H,i}}\, {}^{M_{H,i}}\mathbf{T}^{S_{H,i}}. \tag{S1}$$

The external view estimates ${}^{E}\mathbf{T}_{t}^{D_H}$, and the hand-mounted view estimates ${}^{C_{H,j(i)}}\mathbf{T}_{t}^{M_{H,i}}$. The calibrated transforms ${}^{D_H}\mathbf{T}^{C_{H,j(i)}}$ and ${}^{M_{H,i}}\mathbf{T}^{S_{H,i}}$ describe the camera mounts and the contact module's geometry, respectively, and remain fixed during recording. Calibrated camera intrinsics and target keypoint coordinates specified in physical units make the recovered translations metric.

The four local corners of each contact surface are known from its geometry. For corner ℓ, let $\bar{\mathbf{v}}_{i,\ell} = (v_x, v_y, v_z, 1)^{\top}$ be its homogeneous coordinates in $S_{H,i}$. The demonstration target used for contact-surface retargeting in D3 is

$$\mathbf{p}_{i,\ell,t}^{H} = \left[{}^{E}\mathbf{T}_{t}^{S_{H,i}}\, \bar{\mathbf{v}}_{i,\ell} \right]_{1:3}, \tag{S2}$$

where $[\cdot]_{1:3}$ retains the three Cartesian coordinates.

3) Joint Multi-Tag Pose Estimation: DeepTag provides dense image keypoints on each detected tag. We map their canonical grid locations to 3D points using the tag's known dimensions, decoded orientation, and placement on the rigid target. We aggregate these 3D–2D correspondences across

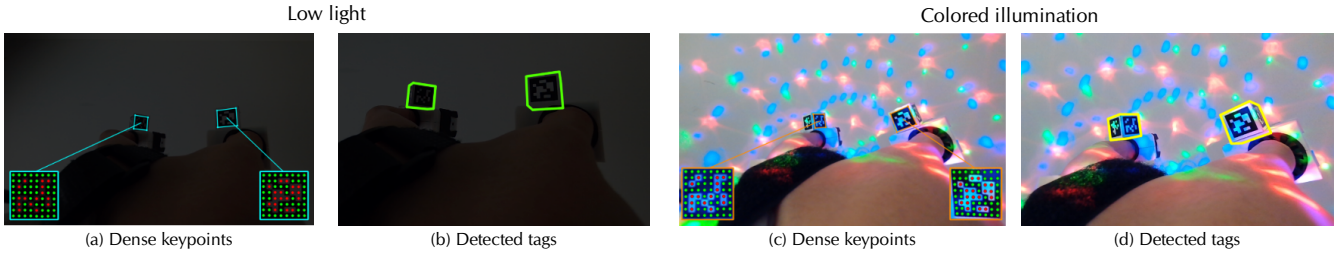


Fig. S2. **Qualitative lighting robustness of fiducial detection.** (a,b) Low-light examples. (c,d) Examples with spatially varying colored illumination. For each condition, the left panel shows dense keypoints with enlarged tag-grid insets, and the right panel shows detected tag outlines. All detection overlays are preserved from the original images.

visible tags and solve for one target pose with multi-tag Perspective-n-Point (PnP) [7].

For a target frame M observed by camera C, let $\mathbf{x}_k$ be a known 3D keypoint in M and $\mathbf{u}_k$ its detected image location. The pose refinement minimizes the joint reprojection error over accepted keypoints $\mathcal{I}$:

$$ {}^{C}\mathbf{T}_{t}^{M} = \underset{\mathbf{T}\in \mathrm{SE}(3)}{\operatorname{argmin}} \sum_{k\in\mathcal{I}} \left\| \pi_{\mathbf{K},\mathbf{d}}\left([\mathbf{T}\bar{\mathbf{x}}_k]_{1:3}\right) - \mathbf{u}_k \right\|_2^2, \tag{S3} $$

where $\bar{\mathbf{x}}_k = (\mathbf{x}_k^\top, 1)^\top$, and $\pi_{\mathbf{K},\mathbf{d}}$ projects a camera-frame point using calibrated intrinsics $\mathbf{K}$ and distortion parameters $\mathbf{d}$.

For observations spanning multiple target faces, the implementation initializes the pose with SQPnP inside RANSAC, rejects inconsistent keypoints and tags, and refines the pose with Levenberg–Marquardt using the retained correspondences. When only one face is visible, we use planar pose estimation with reprojection and face-orientation checks. Joint estimation lets visible tags contribute redundant constraints when other tags are occluded, provided enough valid keypoints remain. The same estimator supplies both tracked transforms in Eq. (S1).

4) Detection under Challenging Illumination: Figure S2 shows qualitative examples of fiducial detection under low light and spatially varying colored illumination. The dense keypoint overlays and detected tag outlines illustrate successful detections under both lighting conditions.

C. Contact-Surface Retargeting

The retargeting implementation optimizes the robot root pose and active finger joints over each recorded sequence. Camera alignment initializes the root pose; it is not an additional objective in the free-wrist variant. Inactive joints remain fixed. The objective combines contact-corner error with first-order temporal regularization.

1) Coordinates and Objective: The numerical optimization expresses both human targets and robot predictions in the moving dorsal fiducial frame D_H, which is rigidly attached to the back of the hand. The estimated transform ${}^{E}\mathbf{T}_{t}^{D_H}$ maps this moving frame into the fixed external tracking-camera frame E; applying the same transform to both targets and predictions leaves their instantaneous Euclidean contact error unchanged. Let $\widetilde{\mathbf{X}}_t = (\widetilde{\mathbf{R}}_t, \widetilde{\mathbf{x}}_t)$ be the robot root pose in this frame, and let $\mathcal{V}$ contain all valid contact-corner tuples (t, i, ℓ). All retained active corners are weighted equally; inactive or invalid corners are omitted. The contact term is

$$ \mathcal{L}_{\text{contact}} = \frac{1}{|\mathcal{V}|} \sum_{(t,i,\ell)\in\mathcal{V}} \left\| \widetilde{\mathbf{p}}_{i,\ell,t}^{R} - \widetilde{\mathbf{p}}_{i,\ell,t}^{H} \right\|_2^2. \tag{S4} $$

Positions are expressed in meters. For consecutive retained frames, define $\delta\mathbf{q}_t = \mathbf{q}_t - \mathbf{q}_{t-1}$, $\delta\mathbf{x}_t = \widetilde{\mathbf{x}}_t - \widetilde{\mathbf{x}}_{t-1}$, and $\delta\boldsymbol{\omega}_t = \log(\widetilde{\mathbf{R}}_{t-1}^\top \widetilde{\mathbf{R}}_t)$, where the rotation logarithm returns a three-dimensional rotation vector. With temporal edge weights e_t and $Z_e = \max(\sum_t e_t, 1)$, the temporal term is

$$ \begin{aligned} \mathcal{R} = \frac{1}{Z_e} \sum_t e_t \Big[& \lambda_q \|\delta\mathbf{q}_t\|_2^2 + \lambda_x \|\delta\mathbf{x}_t\|_2^2 \\ & + \lambda_R \|\delta\boldsymbol{\omega}_t\|_2^2 \Big]. \end{aligned} \tag{S5} $$

Unit conversions and per-dimension normalizations are absorbed into the temporal coefficients. We minimize $\mathcal{L}_{\text{contact}} + \mathcal{R}$ subject to joint limits. Temporal differences of the root pose are measured in the moving dorsal fiducial frame D_H; the final robot trajectory is transformed to the fixed external tracking-camera frame E.

2) Optimizer Configuration: Table S2 summarizes the task-independent configuration. The solver differentiates through robot forward kinematics, while contact-module geometry remains fixed. Separate recordings have no cross-recording temporal edges and are normalized independently.

TABLE S2: Task-independent retargeting optimizer configuration.

Setting	Value
Solver schedule	4,000 projected Adam steps, then up to 1,000 L-BFGS-B steps
Optimization variables	Scaled to $[-1, 1]$
Floating-root bound	± 0.20 m per axis from initialization
Temporal-gap handling	Edge weight $1/g$ for a g-frame interval; no edge across more than 16 missing frames

D. Context-Camera Mount Optimization

For each robot hand, we fit one fixed mounting pose per context camera across the collected demonstrations, holding the retargeted wrist and finger motion fixed. Let $\mathbf{B}_t$ be the robot-palm pose and $\mathbf{H}_{j,t}$ the human pose of camera j, expressed in a common reference frame after handedness conversion. The mount $\mathbf{M}_j \in \mathrm{SE}(3)$ maps the left camera's

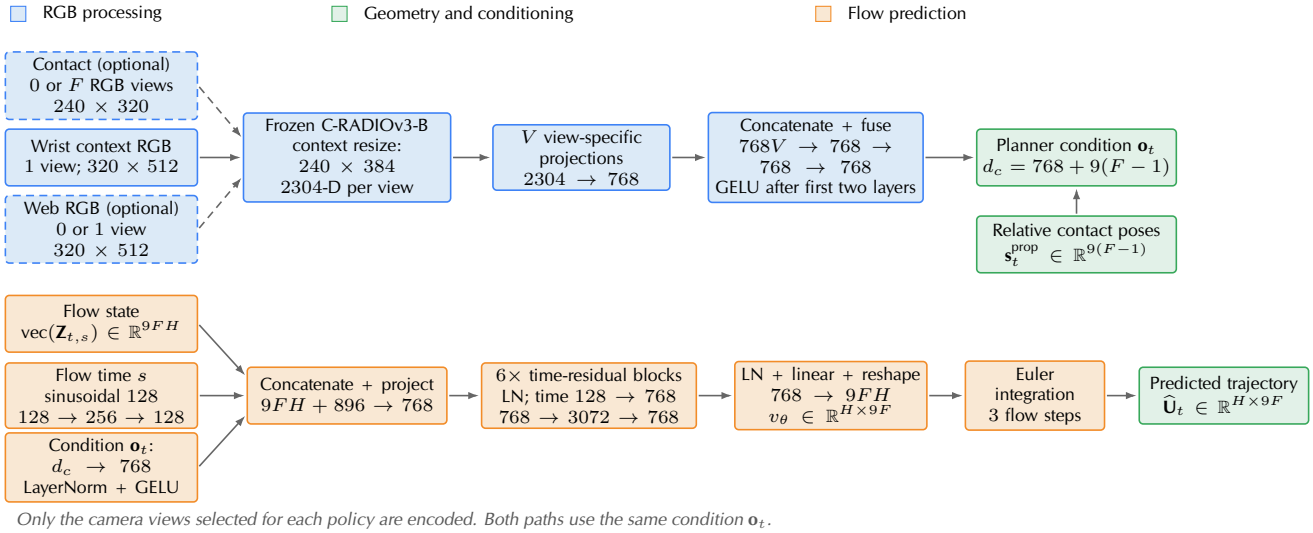


Fig. S3. **Contact-surface planner architecture.** The upper path encodes the V configured RGB views and the current inter-finger geometry into a condition vector. The lower path evaluates the conditional velocity network and integrates it for three Euler steps to predict an H-step contact-surface trajectory. Blue groups RGB processing, green groups geometric inputs, conditioning, and the predicted trajectory, and orange groups flow prediction. Dashed boxes and arrows identify optional camera inputs. Here F is the number of controlled fingers and each contact-surface pose has nine coordinates (three translation and six rotation coordinates).

optical frame into the robot palm frame:

$$\mathbf{M}_j^* = \underset{\mathbf{M}_j}{\operatorname{argmin}} \sum_t w_t \, d_j(\mathbf{B}_t \mathbf{M}_j, \mathbf{H}_{j,t}). \tag{S6}$$

Here d_j penalizes displacement between the predicted and demonstrated camera centers and disagreement between their orientations. The weights w_t give equal total weight to each task, divided equally among its recordings and then their valid frames. The cameras are fitted independently; the LEAP wrist camera additionally constrains its optical $+X$ axis to lie in the palm's XY plane. D4 uses these fitted mounts for viewpoint reprojection; physical mounting calibration is performed separately. Contact-camera placement remains fixed by the shared fingertip modules.

E. Processing Efficiency and Data Quality

Table S3 covers videos to policy-ready features: two contact views and nominal/augmented wrist views, using the augmentation in Section II-A.

TABLE S3: Coin-picking processing on an RTX 4090 D (10 demonstrations, 557 frames). Serial stage wall times include initialization and output writing.

Stage	Time (s)
Visual tracking (image2cube)	57
Contact-surface retargeting (cube2qpos)	43
Viewpoint reprojection and robot overlay	72
Image augmentation and RADIO caching	52
Total	224

Table S4 reports retention among stored task demonstrations, not all collection attempts. Coin rejects one recording shorter than 16 steps; bulb rejects two below 50% valid frames. Invalid frames remain masked during learning.

TABLE S4: Data retention for audited LEAP task datasets. Valid-frame coverage is weighted by recording length, not averaged over recordings.

Task	Retained / stored	Retention	Valid frames
Coin	209 / 210	99.5%	99.3%
Letter	205 / 205	100.0%	95.5%
Bulb	115 / 117	98.3%	79.0%

II. Learning Interface

We first detail the contact-surface planner (L2), then the controller (L3), followed by execution, reuse across tasks, policy-camera preprocessing, and computational cost.

A. Contact-Surface Planner

1) Network Architecture and Dimensions: Figure S3 shows the planner's visual conditioning and flow-prediction paths with their dimensions; Table S5 lists normalization and training settings. Here F is the number of controlled fingers, V the number of RGB views, and H the prediction horizon.

The configured view set defines V: **`CTX`** uses the task's context views, while **`CTX+CC`** additionally includes one contact view per controlled finger. The wrist context view is used for all reported tasks, whereas the web-space context view is enabled only for bulb installation. Separate policies are trained for each view set. Only the selected views are encoded and concatenated; unused cameras are not inputs to that policy.

2) Flow Matching and Conditioning: Given the combined observation $\mathbf{o}_t = (\mathbf{z}_t^{\text{vis}}, \mathbf{s}_t^{\text{prop}})$, the conditional planner $\pi_\theta(\mathbf{U}_t \mid \mathbf{o}_t)$ is implemented with a velocity field v_θ that transforms a source sample $\boldsymbol{\epsilon} \sim \mathcal{U}([-1,1]^{H \times 9F})$ into a contact-motion chunk. For flow time $s \in [0,1]$, generation follows

$$\begin{aligned} \frac{d\mathbf{Z}_{t,s}}{ds} &= v_\theta(\mathbf{Z}_{t,s}, s \mid \mathbf{o}_t), \\ \mathbf{Z}_{t,0} &= \boldsymbol{\epsilon}, \qquad \widehat{\mathbf{U}}_t = \mathbf{Z}_{t,1}. \end{aligned} \tag{S7}$$

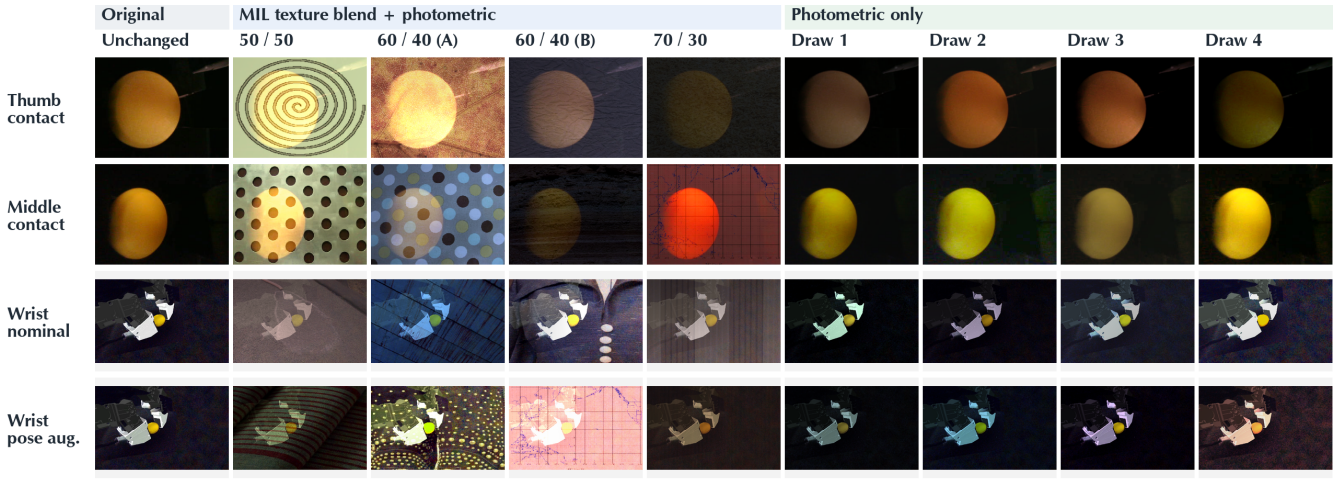


Fig. S4. **Visual augmentation for coin picking.** Rows show the thumb and middle-finger contact views, followed by wrist context views with nominal and perturbed camera extrinsics. The perturbed wrist render also varies robot lighting and materials. Each base image has four whole-image MIL blends with photometric augmentation and four photometric-only variants. Blend columns give original/texture percentages; the two 60/40 candidates use distinct textures.

TABLE S5: Planner normalization and training settings.

Setting	Value
Contact-surface scaling	Translation $\times 20$; six rotation coordinates unchanged
Direct baseline scaling	Global 2nd–98th percentile for non-rotation coordinates; rotation unchanged
Dropout	0
Batch and budget	256; up to 63,000 optimizer updates
AdamW	LR 10^{-4}; $\beta = (0.95, 0.999)$; $\epsilon = 10^{-8}$; weight decay 10^{-6}
LR and EMA schedules	500 warm-up updates then cosine decay; EMA inverse-gamma 1, power 0.75, maximum 0.9999

Here $\mathbf{Z}_{t,s}$ is the evolving chunk, and $\widehat{\mathbf{U}}_t = [\widehat{\mathbf{u}}_{t,1}, \ldots, \widehat{\mathbf{u}}_{t,H}]$ is the predicted relative contact-surface trajectory. We train v_θ with conditional flow matching [8]. For demonstration chunk $\mathbf{U}_t = [\mathbf{u}_{t,1}, \ldots, \mathbf{u}_{t,H}]$, let $\mathbf{U}_{t,s} = (1-s)\boldsymbol{\epsilon} + s\mathbf{U}_t$. Training minimizes

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}\left[\|v_\theta(\mathbf{U}_{t,s}, s \mid \mathbf{o}_t) - (\mathbf{U}_t - \boldsymbol{\epsilon})\|_2^2\right]. \tag{S8}$$

The planner's proprioceptive input consists of non-thumb contact-surface poses relative to the thumb. These poses are computed by forward kinematics from retargeted joints during training and measured robot joints at execution. Only valid future steps contribute to the planner's flow-matching loss; padded horizon entries are masked out. The controller is trained separately, and its parameters remain fixed during planner training.

3) Optimization Settings: LEAP and Wuji share the planner settings in Table S5.

4) Visual Augmentation: Before frozen C-RADIOv3-B encoding, we cache one original and eight augmented images per view. Four blend distinct MIL object or table textures [9] over the full image with original/texture weights of 50/50, 60/40, 60/40, and 70/30, then apply photometric augmentation. Four use photometric augmentation alone. Parameters are sampled independently from Table S6. Figure S4 shows the resulting candidates for a coin-picking demonstration. Context images additionally use camera-extrinsic perturbations and rendered-robot lighting and material variations before image-space augmentation.

Parameter	Range
Temperature τ	$[-0.40, 0.40]$
Tint η	$[-0.12, 0.12]$
Exposure (EV)	$[-1, 1]$
Contrast	$[0.60, 1.40]$
Saturation	$[0.60, 1.40]$
Hue	$[-0.04, 0.04]$
Gamma	$[0.75, 1.35]$
Shading strength a	$[-0.30, 0.30]$
Shading direction φ	$[0, 2\pi]$

TABLE S6: Uniform photometric-augmentation ranges; hue is measured in turns. Temperature and tint set RGB gains to $(e^{\tau-\eta/2}, e^{\eta}, e^{-\tau-\eta/2})$. Shading applies gain $1 + a(x\cos\varphi + y\sin\varphi)/\sqrt{2}$, where $x, y \in [-1, 1]$ are normalized image coordinates.

Training samples uniformly from the nine candidates independently for each view; validation uses originals. The no-visual-augmentation ablation uses only originals while retaining the separate context-rendering selection.

B. Contact-Surface Controller

This section details the controller in L3 of the main paper. Let i index the F controlled fingers, t be the current time, and h be an offset within a trajectory of length H. The robot's current joints are $\mathbf{q}_t$, and $\mathrm{FK}_i(\mathbf{q}_t)$ is contact surface i's pose in the current wrist frame. The predicted $\widehat{\Delta\mathbf{T}}_{i,t,h}$ gives its future pose relative to its current contact-surface frame. Wrist poses are parameterized by translation and the flattened first two rows of the rotation matrix.

1) Geometric Inputs and Command Prediction: The controller C_ϕ in the main paper takes the current joints and the full contact-surface trajectory. It constructs geometric inputs and horizon coordinates internally, then applies the shared network c_ϕ to all predicted steps in parallel. To express the trajectory in the current wrist frame, we compose each relative contact-surface pose with its current forward-kinematics

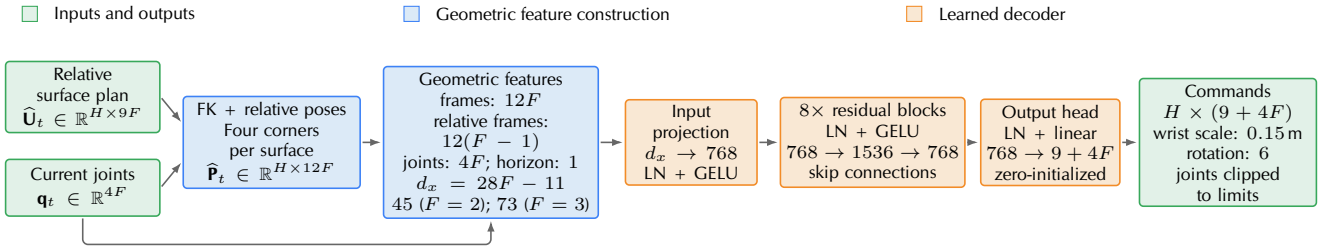


Fig. S5. **Contact-surface controller architecture.** The deterministic geometric front end converts the planner's relative contact-surface trajectory to ordered target corners and constructs absolute and reference-relative frame features. A shared residual MLP then decodes every horizon step in one batched forward pass. Green denotes inputs and outputs, blue denotes deterministic geometric feature construction, and orange denotes the learned decoder. Here F is the number of controlled four-joint fingers, d_x is the per-step feature dimension, and $H = 16$ is the trajectory horizon.

pose. We reconstruct the rotation from its six-dimensional representation and compute each predicted corner as

$$\hat{\mathbf{p}}_{i,\ell,t,h} = \left[\mathrm{FK}_i(\mathbf{q}_t)\widehat{\Delta\mathbf{T}}_{i,t,h}\bar{\mathbf{v}}_{i,\ell}\right]_{1:3}. \tag{S9}$$

Here $\bar{\mathbf{v}}_{i,\ell}$ is the homogeneous local corner defined in Section I-B. The concatenated corners form $\hat{\mathbf{P}}_{t,h} \in \mathbb{R}^{12F}$. A deterministic residual MLP predicts wrist and finger commands for each horizon step:

$$(\widehat{\Delta\mathbf{X}}_{t,h}, \hat{\mathbf{q}}_{t,h}) = c_\phi(\hat{\mathbf{P}}_{t,h}, \mathbf{q}_t, h/H), \tag{S10}$$

where $\widehat{\Delta\mathbf{X}}_{t,h}$ is the future wrist pose relative to the current wrist, and $\hat{\mathbf{q}}_{t,h}$ contains absolute joint targets. The controller derives target-frame and thumb-relative-frame features from the corners and combines them with normalized joints and h/H. Each frame feature has 12 entries: three translation coordinates scaled by 10 and all nine rotation-matrix entries, unlike the planner's nine-coordinate pose representation. The output has $9+4F$ entries for the supported four-joint fingers. Joint targets are clipped to their limits, although this does not guarantee that every predicted multi-surface trajectory is kinematically reachable.

2) Augmented Motion Supervision: We augment wrist and joint trajectories using the endpoint perturbations in Table S7. For horizon step h, the smooth ramp $\alpha_h = s_h^2(3-2s_h)$, with $s_h = h/H$, scales translation and joint offsets and the angle of the left-multiplied rotation perturbation. After clipping joints to their limits, forward kinematics supplies consistent contact-corner targets. Following an action-regression warm-up, we minimize

$$\mathcal{L}_{\mathrm{ctrl}} = \mathbb{E}\left[\frac{1}{4F}\sum_{i,\ell}\left\|100(\mathbf{p}_{i,\ell}^{\mathrm{FK}}(\hat{\mathbf{a}}) - \mathbf{p}_{i,\ell})\right\|_2^2\right] + \lambda_a\mathcal{L}_{\mathrm{act}}, \tag{S11}$$

where $\hat{\mathbf{a}}$ is the predicted wrist–joint action and $\mathbf{p}_{i,\ell}$ is the target contact-surface corner. The factor 100 converts positions from meters to centimeters before squaring the residual. $\mathcal{L}_{\mathrm{act}}$ is normalized wrist–joint regression error, and $\lambda_a = 0.01$. The geometric term directly penalizes realized contact-surface error, while action regression regularizes the kinematic solution.

TABLE S7: Controller motion-augmentation and base optimization settings.

Setting	Value
Motion augmentation	
Endpoint translation	Independent uniform offsets in $[-0.02, 0.02]$ m per axis
Endpoint rotation	Uniform angle in $[-0.15, 0.15]$ rad; uniformly sampled axis
Endpoint joint offsets	Gaussian; standard deviation 8% of each joint's range
Application	Probability 0.9 per batch; independent perturbations per trajectory
Base optimization	
Batch sizes	Training 512; validation 128
Warm-up	1,000 action-only updates; LR 10^{-4}
Main objective	Geometric loss $+0.01\mathcal{L}_{\mathrm{act}}$; LR 3×10^{-6}
AdamW	$\beta = (0.9, 0.99)$; weight decay 10^{-6}
Stabilization	Gradient-norm clip 10; EMA decay 0.999
Budget and validation	At most 20,000 updates; validate every 500 updates
Stopping target	0.1 mm worst-task mean corner error

3) Architecture and Optimization: The residual MLP in Fig. S5 is shared across all horizon steps and evaluates them in parallel. Table S7 lists its motion-augmentation and base optimization settings.

C. Execution and Reuse

The planner predicts $H = 16$ targets at 10 Hz, spanning 1.6 s; execution uses an eight-target chunk (0.8 s at nominal speed). The 10 Hz rate specifies trajectory sampling, not replanning frequency. The controller remains frozen across tasks that share the robot hand, controlled fingers, joint ordering, contact geometry, and trajectory sampling; a different hand or finger set requires its own controller.

D. Discussion on Next-State Action Targets

Following UMI [10], we use future demonstrated poses as action targets, expressed relative to the current pose. These targets describe motion rather than contact forces, but CIMI can still perform contact-rich manipulation without force sensing. Three complementary mechanisms help explain this behavior.

1) Compliant Contact: With two friction-tape layers on the robot and one on the human interface, the extra robot-side thickness can compress against the object at the same

nominal module pose. This tape deformation can produce contact loading without targets inside the object. UMI similarly uses passive fingertip compliance [10].

2) Frictional Support: The friction tape helps resist slip under the normal load supplied by robot actuation. Its role is to support contact, not to measure or regulate force.

3) Relative-Motion Feedback: Replanning anchors predicted motion to the current configuration, allowing visual and proprioceptive feedback to guide renewed corrections toward contact. This is consistent with DexUMI's explanation of relative-action robustness [11].

Future work could add post-contact tactile feedback [11] or augment pose demonstrations with force or wrench measurements for force-informed actions [12, 13].

E. Policy Camera Inputs and Preprocessing

Table S8 specifies the task-dependent camera selections. **`CTX`** includes only the listed context views; **`CTX+CC`** adds the listed contact views while retaining the same context inputs. The dataset may store additional images for reconstruction or visualization; these are not automatically supplied to the policy.

For the two-finger LEAP tasks, the human thumb and middle-finger contact modules map to the robot thumb and index finger; the corresponding camera images follow the same mapping. Bulb uses the human and robot thumb, index, and middle fingers. Each selected stereo unit supplies one RGB policy view, using colors from its rectified left image. Both stereo images are used for depth reconstruction; they are not treated as two separate RGB policy inputs.

The deployment implementation rectifies the live stereo pair, reconstructs a colored point cloud, and rasterizes it into the context image grid with nearest-depth visibility. Stereo reconstruction uses an 800×500 grid, and the policy context raster is 512×320. This preprocessing preserves the rasterization used during training. The policy receives the resulting RGB image, not depth or a point cloud. Human-to-robot viewpoint conversion and robot-geometry overlay are performed on demonstration data offline.

Hand	Task	`CTX`	Added contact views
LEAP	Cube	Wrist	Thumb, index
LEAP	Coin	Wrist	Thumb, index
LEAP	Letter	Wrist	Thumb, index
LEAP	Bulb	Wrist, web-space	Thumb, index, middle
Wuji	Cube, coin	Wrist	Thumb, middle

TABLE S8: Policy camera inputs. Finger names identify the controlled robot fingers.

F. Training and Inference Efficiency

With cached features, the LEAP letter planner takes 24.3 min for 63,000 updates on an RTX 4090 D, including initialization, validation, and checkpointing. Offline processing is reported separately in Table S3.

Table S9 uses the same GPU model: batch one, two contact views, one wrist view, three flow steps, and 16 future targets. RGB-to-command timing includes encoding and feature transfer, but excludes capture, stereo reconstruction, rasterization, and robot communication.

TABLE S9: Inference latency (ms). Both policy paths include the controller; the timings are not additive.

Timed path	Median	95th percentile
Prepared RGB to commands	7.47	7.91
Cached features to commands	2.66	2.80
Controller only (16 targets)	0.84	0.98

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